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MILITARY SPECIFICATION

BUSES, MOTOR: FORWARD CONTROL, 28, 36 AND 44 ADULT PASSENGER; 64 AND 82 SCHOOL AGE PASSENGER, 4x2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel engine driven, forward control type buses. Buses include a forward, mid-mounted or rear-mounted engine, manufacturer's standard rear wheel drive, soft ride suspension, heavy duty bus chassis frame, and single front and single rear axles. Vehicles procured under this specification are commercial items which shall be warranted (by a single prime contractor) as specified in acquisition documents.

* 1.2 Classification. Buses shall be of the types as specified (see 6.2):

Type I - 28 adult passenger
Type II - 36 adult passenger
Type III - 44 adult passenger
* Type IV - 64 school age passenger
* Type V - 82 school age passenger.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2310

AMSC N/A

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* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

A-A-393	- Extinguisher, Fire, Dry Chemical (Hand Portable).
W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).

STANDARDS
FEDERAL

FED-STD-297	- Rustproofing, Commercial (Nontactical) Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1474	- Noise Limits for Army Materiel.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

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(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards
ASTM Designation: A525 - Specifications for General Requirements for Delivery of Zinc-Coated (Galvanized) Iron or or Steel Sheet, Coils and Cut Lengths Coated by the Hot-Dip Method.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SAE, INC.
SAE Standards and Recommended Practices

	J198	- Windshield Wiper Systems - Trucks, Buses, and Multipurpose Vehicles.
	J381	- Windshield Defrosting Systems Test Procedure - Trucks, Buses, and Multipurpose Vehicles.
	J382	- Windshield Defrosting Systems Performance Requirements - Trucks, Buses, and Multipurpose Vehicles.
	J537	- Storage Batteries.
*	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
	J588	- Turn Signal Lamps.

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	J589	- Turn Signal Switch.
*	J683	- Tire Chain Clearance-Trucks , Buses and Combinations of Vehicles (DoD adopted).
	J688	- Truck Ability Prediction Procedure (DoD adopted).
	J816	- Engine Test Code - Spark Ignition and Diesel.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to the SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA
Vehicle Code of California.

(Application for copies of State of California publications should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.9, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented in the bus builder's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

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3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2). When specified (see 6.2), concealed markings shall be furnished. When specified (see 6.2), the exterior color shall be gloss yellow matching color chip No. 13432 of FED-STD-595.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle chassis and underbody including fenders, skirts, and wheelwells, shall be rustproofed in accordance with FED-STD-297.

3.1.1.3 Heavy duty cooling system. A heavy duty cooling system shall be furnished that will maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.1.1.4 Coolant system and indicators. The coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than two-quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.1.1.5 Engine provisions for bus heating. The engine shall have plugged openings for the purpose of supplying hot water for the bus heating system. The openings shall be suitable for attaching 3/4 inch pipe thread/hose connector(s). The engine shall be capable of supplying water having a temperature of at least 170°F at 0°F ambient temperature at a flow rate of 6 gallons per minute at the return end of 30 feet of one inch inside diameter (i.d.) automotive hot water heater hose.

3.1.1.6 Wood treatment. Wood flooring (see 3.5.3) shall be treated in accordance with MIL-STD-1223.

3.1.1.7 Rear towing devices. Two closed loops for towing the bus shall be provided on the rear of the bus. The towing loops at the rear of the bus shall withstand, without any signs of failure, lifting the rear end of the bus, turning the bus within its minimum turning radius, and backing the bus on a level roadway. Decals shall be provided on front and rear bumpers stating "LIFT AND TOW FROM REAR OF BUS." Complete towing instructions shall be furnished with the operator's manual.

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3.1.1.8 Front tow attachments. Two tow chain attachment devices shall be provided at the front of the bus to allow recovery of the bus from a ditch or other impediment. Each front attachment device shall provide an ultimate strength at least equal to the GVW rating of the vehicle furnished.

3.1.1.9 Towing brakes. Unless otherwise specified (see 6.2), in addition to the brake requirements specified herein (see 3.4.11), an emergency system for controlling at least the front brakes of the bus from a towing vehicle shall be provided. The installation shall be complete with air brake couplers; relay emergency valve with no-bleed-back feature, (except when spring applied emergency brake is furnished); and additional air lines and fittings. Towing brake installation shall not compromise conformance to any Federal Motor Carrier Safety Regulation or Federal Motor Vehicle Safety Standard (FMVSS) referenced herein. Couplers shall be mounted in a protected location at the rear of the bus and shall permit ready attachment of air hoses from the towing vehicle. Service coupler shall be located on the curbside and emergency coupler on the streetside.

3.2 General design.

* 3.2.1 Federal Motor Vehicle Safety Standards. Vehicles and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture, including standards applicable to only school buses, except:

(a) Exterior color shall be as specified in 3.1.1.1.
(b) School bus signal lamps may be omitted when specified for types I, II or III adult passenger buses (see 3.4.2.6.4).
(c) Seats shall be in accordance with 3.5.8 in lieu of FMVSS No. 222 for types I, II and III adult passenger buses.

3.2.2 Air pollution control. The vehicle shall comply with Environmental Protection Agency Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level at the operator's location with driver in seated position shall be not greater than 85dB(A) when measured in accordance with MIL-STD-1474. The vehicle exterior sound level shall conform to the Interstate Motor Carrier Noise Emission Standards of the Environmental Protection Agency when tested in accordance with the Regulations of the Department of Transportation, Part 325. When specified (see 6.2), the exterior sound level shall not exceed the noise limits established by the California Vehicle Code, Section 27200.

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3.2.4 Curb weight. Curb weight shall include the weight of the chassis with all attachments, accessories and equipment; full complement of fuel, lubricants and coolants.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall include the curb weight, the driver weight at 175 pounds and the passenger payload weight evenly distributed through the passenger seat and standee area.

3.2.5.1 Passenger payload. The passenger payload weight shall be the weight of the specified number of passengers computed at 175 pounds each plus 25 percent of this weight as an allowance for standee and carry-on baggage for types I, II and III adult passenger buses and shall be the weight of the specified number of passengers computed at 120 pounds each for types III and IV school age passenger buses.

3.2.6 Dimensions. Bus dimensions shall be as follows:

(a) Overall width shall be not greater than 96 inches. Width over tires shall not be greater than 100 inches.
(b) Overall height shall be not greater than 125 inches for vehicles without air conditioning and not greater than 140 inches for vehicles with air conditioning.
* (c) Overall length shall be not greater than 30-1/2 feet for type I bus, not greater than 35-1/2 feet for type II bus, not greater than 38 feet for type III and type IV buses, not greater than 40 feet for type V bus, and shall include all front and rear protrusions.
* (d) When the bus is loaded to its specified passenger payload (see 3.2.5.1), the ground clearance shall be not less than 8-1/2 inches under axles and not less than 10 inches under all other parts.

3.2.7 Ratings. Vehicle ratings shall be manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

3.2.8 Accessibility. The design of the vehicle and optional equipment shall permit ready accessibility for routine servicing and shall permit accessibility for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the bus loaded to specified GVW.

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* 3.3.1.1 High speed gradeability. At 55 miles per hour (mph) the vehicle shall ascend a grade of not less than 1.5 percent for types I and II, 1.2 percent for types III and IV, and 0.9 percent for type V bus. Gradeability requirements shall be met with the transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed not greater than 4 mph.

3.3.2 Service brakes. Service brakes shall control and hold the vehicle, when loaded to its specified gross weight, on a 30 percent grade. Under all conditions of loading, the service brakes shall stop the bus, from a speed of 20 mph, within a distance of 35 feet from the point at which movement of the brake pedal begins. Deceleration and stopping distance shall be demonstrated on dry, hard, approximately level road surface that is free from loose material.

* 3.3.3 Turning radius. The turning radius, defined as the distance from the turning center to the centerline of the tire describing the largest circle while the vehicle is executing its sharpest practicable turn, shall be not greater than 34 feet for type I, 40 feet for types II, III and IV and 42 feet for type V bus.

3.4 Chassis.

3.4.1 Engine. The engine furnished shall be manufacturer's standard or optional diesel engine for the commercial model vehicle which meets or exceeds the requirements of this specification. Engine net torque rating shall be not less than 414 pound feet. The engine shall be mounted within the body at the front of the vehicle, shall be mounted between front and rear axles under the floor, or shall be mounted within the body at the rear end of the vehicle. The diesel engine shall be liquid cooled, compression ignition, two-stroke or four-stroke cycle, with not less than six cylinders. Engine net torque and net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349. The engine model furnished shall have been in fleet usage for at least one year prior to contract award. The engine furnished shall be as recommended by the engine manufacturer for the vehicle model furnished.

3.4.1.1 Engine compartment. Engine compartment ventilation and design shall allow operation without damage to engine, transmission and all components when vehicle is operated in ambient temperatures of up to 120°F. Ready access to engine compartment shall be provided for servicing and routine maintenance of engine and engine components. Hinged engine compartment covers or doors shall be provided with a device to hold them open for servicing and maintenance.

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3.4.1.2 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.3 Governor. An engine governor shall be furnished and set to limit engine speed to maximum recommended operating speed. The governor shall be sealed at the specified setting.

3.4.1.4 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided to maintain engine temperature within the engine manufacturer's recommended operating range. Control shall include complete thermostatic control of all coolant flow through the radiator, or partial thermostatic control of coolant flow through the radiator, and thermostatically controlled radiator shutters. When power plant heaters are specified (see 3.4.1.5), thermostatically controlled radiator shutters shall be provided.

3.4.1.5 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, engine oil heater, and fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be furnished when power plant heaters are specified. When power plant heaters are provided, thermostatically controlled radiator shutters specified in 3.4.1.4 shall be provided. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. A carrier for the connecting cable shall be mounted within the engine compartment or stowage compartment(s) and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed;
(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening;

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(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure to ambient air temperatures as low as minus 60°F and shall include a thermostat to limit the temperature of the electrolyte to not more than 80°F;

(d) A fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.1.6 Fan clutch. A fan clutch shall be furnished. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling.

3.4.1.7 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.8 Fuel fired engine preheater. When specified (see 6.2), a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and a circulating pump; and shall be connected directly across the engine coolant inlet and outlet. The heater shall be capable of starting and operating at minus 60°F and shall heat the engine to plus 40°F from minus 60°F in not more than one hour. The system shall be equipped with a light visible to the driver, to indicate that the preheater is operating.

3.4.2 Electrical systems. Electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

3.4.2.1 Radio interference suppression. Vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radio interference shall be suppressed to the same level as the vehicle chassis.

* 3.4.2.2 Diesel engine starting system. A 12- or 24-volt starting system, with 12-volt lighting system shall be furnished. Engine starting equipment shall include an ether starting system or a glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

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3.4.2.3 Alternator. The rated amperage capacity of the alternator provided shall be not less than 130 amperes at not more than 60 percent of engine governed speed and not less than 40 amperes at engine idle speed. Transistor type voltage regulator shall be furnished.

3.4.2.4 Safety switch. An engine starting device and a safety switch shall be provided in the engine compartment on mid-mounted and rear-mounted engine buses so arranged that the engine cannot be cranked from the front end with the switch in the "off" position. Means shall be provided to prohibit cranking the engine using the engine compartment starting device except when the transmission is in neutral position.

* 3.4.2.5 Batteries. Batteries shall be of 12-volt potential. Batteries shall have a total reserve capacity rating of not less than 480 minutes and a cold cranking rating at 0°F of not less than 1740 amperes, both ratings in accordance with SAE J537. Batteries shall be of the maintenance free type, having the maintenance free characteristics listed in W-B-131. When a mid-mounted or rear-mounted engine is furnished, batteries may be mounted in the engine compartment. When mounted in the engine compartment, batteries shall be mounted in an accessible location so that all cells are exposed for inspection and servicing. The batteries shall not be exposed to excessive engine heat or exhaust heat. A sliding tray shall support the batteries in the pulled out position and, in this position, shall provide for ease of battery inspection, maintenance and replacement. Trays shall be coated with acid-resistant material. A battery access door shall be provided on the side of the body.

3.4.2.6 Lighting. All vehicle (chassis and body) lights, reflectors, and wiring shall be as specified herein, and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26. The electrical system shall be 12-volt. Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) or vehicle bumpers. An illuminated rear license plate holder shall be provided. The bus shall be equipped with the manufacturer's standard foglights. Two backup lights shall be provided at the rear of the vehicle.

3.4.2.6.1 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 class A and shall be mounted on the steering column. The vehicle shall be provided with single-faced front signal units and with single-faced rear signal units installed in accordance with SAE J588. Operating units shall be provided with a visible and audible type flash indicator. Turn signals shall be of the self-cancelling type. Additional armored type amber turn signals, one mounted on each side of the bus above or near the front wheelwell, shall be furnished.

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3.4.2.6.2 Passenger's signal buzzer. When specified (see 6.2), the manufacturer's standard signal buzzer equipment shall be furnished, operated by suitable pull cords on both sides of bus, and provided with a driver's cutout switch.

3.4.2.6.3 Interior lighting. Interior lighting shall include 15 candle-power domelights, of a grade and quality providing not less than 12 square inches of polished reflecting surface fitted with frosted translucent lens providing not less than 30 square inches of area. Domelights shall be installed at a height at least above the window line on each side of the bus, approximately adjacent to the first or second seat from the front, and thereafter over each second seat. One domelight shall be installed over the driver's area and controlled with a separate switch. Fluorescent lamps and fixtures may be furnished in lieu of the domelights. Fluorescent lamps shall be located on the longitudinal centerline of the aisle of the passenger area for the entire length. There shall be not less than 4 lamp fixtures in the row. Lamps shall be not less than 40-watt, and of the high output capacity type. Lamps shall provide a broad distribution of light. Lamps shall be white. A non-glare deflector or shade and tube guard shall be provided. The fluorescent lamps shall be furnished with inductive capacitive type filters for each lamp to reduce radio interference (see 3.4.2.1). Door-operated ceiling lamp(s) and step lamp(s) shall be furnished and installed to provide illumination of stepwell and ground adjacent to the steps. An emergency door identification red light with not less than a 3-inch diameter lens shall be provided at each emergency exit in accordance with Federal Motor Carrier Safety Regulation 393.92 (see 3.5.5.2). A signal audible to the driver shall be provided to indicate when rear and emergency door(s) are not properly secured during operation of the vehicle. Wiring shall be accessible for replacement.

* 3.4.2.6.4 School bus red signal lamps. Types III and IV school age passenger buses and unless otherwise specified types I, II and III adult passenger (see 6.2), buses shall be equipped with four red signal lamps conforming to Federal Motor Vehicle Safety Standard No. 108.

3.4.2.7 Master cutoff switch. When specified (see 6.2), an ignition key or driver manually operated master switch shall be provided to control a magnetic solenoid to shut off all body electrical equipment except: (a) headlights (b) running lights required by FMVSS 108 and (c) the starting motor. The switch shall be identified as to function. The solenoid shall be rated for continuous full load.

3.4.3 Fuel system. Fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

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* 3.4.3.1 Air cleaner. A single-stage or two-stage dry type air cleaner shall be furnished. The air cleaner shall include a progressive type indicator to display the degree of air restriction. The indicator shall be located adjacent to the air cleaner and shall have a red line or other clear mark to indicate when the filter should be changed.

* 3.4.3.2 Fuel tank(s). Safety fuel tank(s) conforming to Federal Motor Carrier Safety Regulations 393.65 and 393.67 shall be furnished. Total capacity of the fuel tank(s) shall be not less than 60 gallons for vehicles without air conditioning and not less than 80 gallons for vehicles with air conditioning. When more than one tank is furnished to feed the main diesel engine, means shall be provided to assure equalized fuel level in both tanks and means shall be provided to monitor the fuel level of either tank from a single fuel gage. Tanks shall be located under the floor of the bus. A filler shall be arranged for safe and convenient filling from the exterior of the body. The access for the filler neck shall be designed to prevent the entry of spilled fuel to any point inside the passenger compartment or entrapment between body walls. Sealing may be accomplished by use of a neoprene rubber thimble or grommet, or other device installed in access hole. When tank filler cap is located behind body skirting, a front hinged access door shall be furnished. A photoetched, corrosion-resistant plate, permanently protected decal or stencil, shall be located in a conspicuous place near the tank filler cap and shall state "diesel fuel" and "main engine" or "main and auxiliary engine" as appropriate. When an air conditioning system is furnished (see 3.5.13), the air conditioner auxiliary diesel engine may operate from the main engine fuel tank(s) or from a separate tank specified in 3.5.13.2.6.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83, shall be substantially constructed, effectively leakproof, and quiet in operation. A properly supported exhaust tailpipe shall be located to properly and effectively minimize the possibility of exhaust fumes entering the passenger area under all operating conditions. The exhaust system shall include muffler(s).

* 3.4.5 Transmission. Transmission input torque capacity shall be at least equal to the maximum torque delivered by the engine. Transmission and axle gear ratio shall be selected to provide the performance specified in 3.3.1 through 3.3.1.2.

3.4.5.1 Automatic or semiautomatic transmission. Unless otherwise specified (see 3.4.5.2), an automatic or semiautomatic transmission shall be provided. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. The normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. When specified (see 6.2), the automatic transmission shall have a lockup in the two highest forward speeds.

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3.4.5.2 Manual transmission. When specified (see 6.2), the manufacturer's standard manual synchromesh transmission shall be furnished. Transmission shall provide not less than five forward speeds and one reverse speed.

3.4.5.2.1 Clutch. The clutch shall be the largest capacity clutch offered for the size of engine furnished with clutch torque capacity exceeding maximum delivered engine torque.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction. Propeller shaft guards shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.89.

3.4.7 Chassis frame. The chassis frame shall be of the forward driver control, heavy duty type. The frame shall be continuous from bumper to bumper or shall be continuous and of the same cross-section at least from the rear bumper to the forward front suspension hanger bracket. When continuous frame rails are furnished, the section modulus and resisting bending moment of the frame rails shall be as recommended by the chassis and body builder(s) to support the body and specified load under severe operating conditions. When a non-continuous frame rail is provided, the front rail section shall be reversed to provide clearance for mounting a diesel engine between the frame rails; the front and rear rail sections shall be overlapped and joined together by both bolted and welded construction; and the resisting bending moment shall be not less than 640,000 inch-pounds for the main rail section and not less than 390,000 inch-pounds for the front rail section. Bus chassis shall be built by the bus body builder or to the bus body builder's specification for the specific model being furnished. Chassis frame, body, and undercarriage shall be so designed and constructed as to support body and load to maintain stability under severe operating conditions.

* 3.4.8 Suspension. The vehicle shall be equipped with bus manufacturer's standard or optional system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. The vehicle shall be provided with spring, full air or combination spring and air suspension on the front axle. The rear suspension shall be full air or a rectilinear shear rubber system. The vehicle shall be provided with hydraulic double-acting shock absorbers, not less than two on the front and two on the rear suspension. The suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the vehicle from the applicable jacking location. Air systems shall incorporate leveling valve(s) with time delay to avoid

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constant air consumption. Shock absorbers, airbags, and solid rubber springs shall be fabricated of materials capable of operating in ambient temperatures from minus 50°F to plus 125°F. The suspension system shall be recommended and warranted by the chassis supplier to meet specified conditions under mobile operating conditions.

3.4.8.1 Front suspension. The spring, full air, or combination spring and air suspension design shall be such that the natural frequency of vertical vibration at the front wheels under all conditions of loading shall not exceed 2.0 cycles per second. Torque rods shall be provided if available as manufacturer's standard or option to increase longitudinal and lateral stability.

* 3.4.8.2 Full air rear suspension. The rear axle equipped with full air suspension shall have two or four air springs of adequate diameter. The suspension design shall be such that the natural frequency of vertical vibration about any transverse axis shall not exceed 1.5 cycles per second at rated capacity for specified GVW. Full air system shall have none of the spring weight carried on anything other than air springs. No other spring medium such as leaf, coil, or rubber springs may be used in conjunction with the air springs. The two-air-spring style system shall include shock absorbers; two one-piece beams and two air springs to transfer the load directly to the axle with the forward portion of the one-piece beam free to move in low-friction wear pads; and a height-control valve system capable of maintaining a given frame height. The full air system shall incorporate four radius rods for the four-air-spring and three radius rods for the two-air-spring type to control lateral and longitudinal movement. Radius rods shall transmit driving and braking forces to chassis and body. Each end of the radius rods shall be equipped with rubber bushings that do not require periodic lubrication. The suspension system shall be so designed that the vehicle can be lifted and towed from the rear without damage to the suspension.

3.4.8.3 Rectilinear shear rubber rear suspension. The rectilinear shear rubber suspension system for the rear axle shall be such that at rated capacity at specified GVW, deflection shall be 3-1/4 inches plus or minus 1 inch. Spring centers shall exceed 50 inches. The system shall be the friction-free type with no lubrication points. Shock absorbers shall be located on not less than 66-inch lateral centers. The spring weight shall be suspended by solid rubber springs, fabricated of either natural rubber or natural-synthetic compound. A stabilizer bar or torque rod shall be provided for increase in longitudinal and lateral stability.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the vehicle is loaded to specified GVW. The vehicle shall be equipped with single front and

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single rear axles. A single reduction drive axle shall be furnished at the rear. Torque rods shall be furnished for the rear axle if available as the manufacturer's standard or option. Axle gear ratio furnished shall enable vehicle performance required in 3.3.1 through 3.3.1.2.

3.4.9.1 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, and tires. The vehicle shall be equipped with single front and dual rear wheels. Rim size, tire size, type, and ply rating shall be the same for all wheels on each vehicle. Tire and rim ratings shall conform to Tire and Rim Association recommendations, for the type and size of tires furnished. When specified (see 6.2), 10 stud disc type wheels shall be furnished.

3.4.10.1 Tires. Tires shall be steel belted radial or when specified (see 6.2), bias ply. Tube or tubeless tires with highway tread shall be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded with specified payload. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. When tube type tires are furnished, inner tubes shall be heavy duty type, and shall be the proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Balancing. Each tire shall balance within practicable limits. Wheels, hubs, and brake drums shall be effectively in balance. Balancing shall be adequate to preclude wheel shimmy at all vehicle speeds.

3.4.10.4 Spare wheel assembly. When specified (see 6.2), a spare wheel or rim shall be mounted on a winch type carrier and installed in a readily accessible location. The spare wheel assembly shall be so located as not to protrude beyond the body. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.16. Threaded fasteners used to secure the spare tire in the carrier shall be constructed of or plated with corrosion resistant material.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The assembly shall include the winch

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type carrier, the spare wheel or rim specified in 3.4.10.4 and a mounted and inflated spare tire. The spare tire shall be of the same size, type, tread design and ply rating furnished on the vehicle.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42, and 393.44 through 393.52.

3.4.11.1 Service brakes. The vehicle shall be equipped with full air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm).
(b) Air storage reservoir(s) with a total capacity of not less than 2240 cubic inches for bus with a spring suspension system and 3600 cubic inches for bus with an air suspension system; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
(c) Foot control, suspended or treadle type.
(d) Air control valves.
(e) Air pressure gage, visible to the driver.
(f) Low air pressure warning, visible and audible.
(g) Service brake stop lamp switch.
(h) Automatic moisture ejector.
(i) Relay emergency valve.
(j) Automatic slack adjusters.

* 3.4.11.2 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.12 Horn. An electric horn shall be provided. In addition, when specified (see 6.2), an air operated horn shall be provided.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual air operated windshield wipers. The wiper system furnished shall conform to SAE J198 and shall sweep the maximum practicable area of windshield surface. Windshield washers shall be

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provided. Washers shall be electric, vacuum or air operated, with control on instrument panel.

3.4.15 Bumpers. A channel type front bumper shall be provided. The rear bumper shall conform to Federal Motor Carrier Safety Regulation 393.86 and shall be as specified in 3.5.11.6. The top of the front bumper shall be provided with a skid-resistant finish to assist personnel in window cleaning (see 3.4.18).

3.4.16 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging mounted tire assembly with a spare assembly (see 3.4.10.5), and shall include at least a hydraulic jack, jack handle, and wheel nut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of loaded vehicle to height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Controls and operating mechanisms. Unless otherwise specified (see 6.2), controls and operating mechanisms and doors shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver when in a seated position. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's seat. Instruments shall be panel mounted and shall be clearly visible to the driver when seated in normal driving position. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing.

* 3.4.18 Windshield cleaning grab handles. A handle shall be furnished at each side of the windshield to provide a handhold for window cleaning while standing on bumper (see 3.4.15). The handle shall be tubular metal or reinforced plastic with a tubular shape.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch.
(b) Ammeter, charging indicator, or voltmeter.
(c) Fuel gage.
(d) Oil pressure gage or indicator.
(e) Engine temperature gage or indicator.
(f) Speedometer.
(g) Recording odometer.
(h) Tachometer.

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(i) Indicator light to show headlight high beam operating.
(j) Hand throttle.

3.5 Body. The body shall be mounted on a chassis or may be of unit body and chassis construction. The bus shall be the transit forward control type with seating for the driver forward of the front axle. The bus shall provide adequate seating for adult or school age passengers specified (see 1.2). The body shall have sides of the full-skirted type, a flat rather than a stepped front design, shall present a generally symmetrical appearance, shall effectively exclude engine fumes and exhaust gases from the bus interior, and shall provide safe and comfortable transportation for passengers and driver.

3.5.1 Inside body dimensions. At seat level, body inside width shall be not less than 90 inches. Between the first and last vertical roof bows, through the center of the aisle, the inside height shall be not less than 76 inches.

3.5.1.1 Front entrance clearance (front mounted engine only). The clear opening in the stepwell, in the entrance area and between the curbside modesty panel and the engine housing, shall be not less than 18 inches.

3.5.2 Construction. Body and roof framing shall be constructed of steel sections, with metal panels attached outside and inside and properly insulated. All parts, where applicable, shall be fastened together in a manner which will preclude loosening of bolts, screws and rivets, and cracking of welded joints when the vehicle is properly maintained and operated under the intended service conditions. Self-tapping screws shall not be used where subjected to stress. Wood shall not be used for structural framing. Outside body panels and other sheet metal shall be of aluminum alloy or steel and of proper gages, not less than the gages, as applicable, regularly furnished by the body manufacturer in buses of the type offered. The body, roof and panel joints shall be watertight and leakproof. Unless zinc coated steel panels are used, joints shall be sealed with "Alumelastic" compound or equivalent. All steel 12 gage (0.1046 inch) and lighter shall be zinc coated in accordance with ASTM Designation A525 and table I, Coating Class G60.

3.5.2.1 Sound level - passenger locations. The octave band sound pressure level average shall not exceed those levels shown in column A of table I. Sound readings shall be taken at three positions in the aisle, (a) next to the front passenger seat, (b) next to the center passenger seat, and (c) next to the rear passenger seat with vehicle traveling at 30 mph in direct drive on a smooth level road with all powered accessories except heater blower (and air conditioner, when furnished) operating and no more than four people within the vehicle.

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At authorized maximum speed for the vehicle and at the noisiest position in the vehicle, the octave band sound pressure levels shall not exceed those shown in column B of table I.

TABLE I. Acceptable noise level.

Octave band limits (cps)	Sound pressure level db, re 0.0002 microbar	
	Column A	Column B
75 - 150	84	--
150 - 300	80.5	92
300 - 600	72	89
600 - 1200	71	87
1200 - 2400	68	85
2400 - 4800	66	84

3.5.3 Floor. Floor frame and floor plates, with entrance step and wheelhousings, shall be of substantial construction properly integrated with the body structure, and shall conform to Federal Motor Carrier Safety Regulation 393.84. Wood parts, including plywood, shall be treated in accordance with 3.1.1.6. Plywood used in floor panels shall be water-resistant, exterior grade, fir or dense southern pine. Floor panels shall be of not less than (a) 3/4-inch, seven-ply plywood, (b) 1/2-inch, five-ply plywood over 10 gage steel; or (c) 5/8-inch, five-ply plywood over 14 gage steel. Steel channels, not less than 2 inches in depth and spaced to provide support for the loads imposed, shall be furnished under the floor.

3.5.3.1 Stepwell. Stepwell(s) shall be enclosed, with two or three steps of uniform height, and constructed of steel covered with 3/16-inch thick rubber or synthetic rubber blend ribbed material. The height of the bottom step shall not exceed 16 inches.

3.5.3.2 Wheelhousings. Front and rear wheelhousings shall include splash shields under the body. Splash shields shall extend around the top of the wheel and to the bottom of the body side skirting, front and rear. Ample clearance for the use of tire chains on drive wheels, or at least on outside dual drive wheels, shall be provided in accordance with SAE J683. Molded rubber splash trim shall be installed around the outside edge of both the front and rear wheelhousings.

3.5.3.3 Floor coverings. The underseat floor area and the wheelhousings shall be covered with not less than 1/8-inch thick rubber or synthetic rubber blend material. The aisle and the entrance area shall be covered with not less than 3/16-inch thick rubber or synthetic rubber blend ribbed material. Coverings shall be properly installed and adequately sealed at the joints. The floor in the driver's area shall be covered with a mat or with the same materials as used in the underseat or aisle area.

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3.5.4 Sides and roof. Sides and roof shall have sheet metal outer panels, lined on the inside and integrated with the body structure.

3.5.4.1 Exterior. Sides shall be fullskirted and equipped with at least two full-length rub rails extending along each side of the body one below the platform level and one approximately at seat height. A substantial snow rail at the bottom of the skirt will be acceptable in lieu of a rub rail if such rub rail is not offered by the manufacturer as a standard or optional item.

3.5.4.2 Interior. Body interior, including sides, roof, and front and rear ends, shall be fully lined with sheet metal panels.

3.5.4.3 Insulation. Nonsettling, nonhygroscopic, fungus resistant, fireproof insulation shall be installed in roof, side, and end walls. In addition, a coat of material shall be applied to the inner (not exposed) surfaces of exterior panels and other exterior sheet metal as added insulation and to prevent drumming. Insulation shall be of sufficient density to enable the air conditioner to meet the requirements of 3.5.13.1.

3.5.4.4 Destination sign compartment. When specified (see 6.2), a destination sign compartment shall be installed at the front center of the roof above the windshield. The compartment shall provide not less than 180 square inches of glass enclosed frontal area, of a height to permit reading of 6-inch letters on the sign roller, and shall be readily accessible, through latched door, from inside the bus. Destination sign shall be a roller type of a capacity to apply 15 listings of 6-inch high letters and shall be operable from the interior of bus. Unless otherwise specified, the destination sign roll will be furnished blank. The compartment shall be illuminated to permit reading of the destination sign at night.

3.5.5 Doors. A service door shall be provided at the front, forward of the front axle, on the side opposite the driving controls, for use by entering and departing passengers. An emergency door or exit shall be provided at the center rear or at the sides of the bus in accordance with Federal Motor Vehicle Safety Standards for school buses. When specified (see 6.2), a hasp for padlocking the service door and a locking bolt for the emergency door or exit shall be furnished. The locking bolt for the emergency door or exit shall include an interlock that prevents the engine from starting when the emergency door is locked. All doors shall be provided with compression or overlapping seals to minimize the entry of dust, cold air, rain, and road wash.

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3.5.5.1 Service door. The service door shall be of the folding leaf type or one-piece type with emergency exit features operable by school children. The service door shall have safety glass panels in upper and lower sections. The door shall have manual or air operated controls for driver's operation. Operating linkage shall be installed within the body or enclosed to preclude accumulation of mud, ice and snow. Minimum dimensions of clear opening shall be 24 by 74 inches, unobstructed by door seals or door opening mechanism.

3.5.5.2 Emergency door or exit. Emergency door(s) or exit(s) shall be of the sedan type equipped with a slide-bar, cam operated latch, and with readily operable interior handle. A permanently installed handle shall be provided for opening the emergency door or exit from outside the bus. A switch operated by the door bolt shall be connected to a buzzer on the instrument panel and a buzzer at the emergency door to indicate to the driver when the emergency door or exit is not safely latched. A red emergency door identification light (see 3.4.2.6.3) shall be installed on the emergency door or exit header(s). The red warning light shall actuate when the ignition switch is in the "on" position. The header(s) shall be marked with the following block letters not less than 2 inches high: "EMERGENCY DOOR" or "EMERGENCY EXIT."

3.5.6 Windshield. A windshield affording maximum practicable driver vision shall be installed at a slope sufficient to minimize external glare or reflections from inside the body.

3.5.7 Windows. Windows shall be of a size and arrangement suitable for the intended service and shall provide maximum practicable vision. Safety glass shall be used in all windows. Windows shall be provided with finger-tip control, permitting ready opening and adjustment to various desired openings. Windows shall be rattleproof and, when closed, rainproof, waterproof and windproof. All windows shall be fitted with locking devices. Driver's window(s) shall permit use for signaling and for ventilation. The top edge of the glass area in the side windows or transom windows, if furnished, shall be not less than 57 inches above the aisle floor level.

* 3.5.7.1 Tinting. When air conditioning is furnished, see 3.5.13 tinted, heat absorbing, safety window glass shall be furnished. Entrance door windows, driver's left side window(s) and windshield shall be in accordance with the manufacturer's standard for tinted glass. Side windows in the passenger area shall be tinted to double density allowing not more than 55 percent light transmission. Rear window(s) shall be tinted to quadruple density allowing not more than 40 percent light transmission. For passenger area side and rear windows, safety tinted glass shall be used.

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* 3.5.8 Seating. Seating shall be provided to accommodate the specified number of passengers and the driver. All seats shall be of the forward facing type, secured to prevent shifting under all operating conditions and shall conform to Federal Motor Carrier Safety Regulation 393.91. The seats shall be constructed, upholstered, covered and arranged as specified herein. Additionally, except for the rearmost seat(s) the top and at least the upper 6 inches of the rear surface of each two-passenger or three-passenger seat back shall be constructed of or covered with force distributing material. The material shall minimize force and spread the area of contact upon impact by an occupant in the next seat to the rear. The rear surface of the seat back below the force distributing material shall be covered with patterned aluminum, stainless steel, aluminized steel or bright finish zinc coated steel.

* 3.5.8.1 Passenger seat covering. All passenger seat and force distributing cover material shall be first grade, heavy duty, plastic-coated, grained surface fabric of not less than 38 ounce weight per linear yard of 54-inch width. All seat materials shall be in accordance with the flameproof requirements of Federal Motor Safety Standard No. 302. Color of upholstery shall be appropriate with exterior color and shall harmonize with interior colors.

* 3.5.8.2 Passenger seating for types I, II and III buses. Transit or suburban-type, nonreclining seating shall be provided for types I, II and III buses to accommodate the specified number of adult passengers. All passenger seats shall be 2-passenger, forward facing type mounted on each side of center aisle. Seats shall be on minimum 36-inch centers for 28- and 36-passenger buses and 29-inch centers for 44-passenger buses. Seat back of rear seats shall be installed at the same angle of inclination as the regular two-passenger seats. Front seat cushions shall be spaced at least 10 inches from modesty panel and 9 inches from back of driver's seat, with the driver's seat adjusted approximately 17 inches from the seat back to the steering wheel and the driver's seat back in the medium tilt or full tilt position. Seats located at or near wheelhousings shall be installed to minimize interference with passenger comfort. Seats at side emergency door(s) shall be spaced to provide 24 inches clearance between seat back and door post. Seats at rear emergency door shall provide a 24-inch wide clearance to emergency door. The 2-passenger seats shall be American Seating Company Model 6521 or equal.

* 3.5.8.2.1 Adult passenger seat construction. Frames of adult passenger seats shall be constructed of formed metal tubing and shall include armrest. Seats shall be manufacturer's standard nominal 36-inch width, except width of rearmost seats and of front seat at entrance may be reduced to a nominal 33 inches, if required to provide adequate access to entrance and emergency exit. Seat cushions shall have a free depth of not less than 17 inches from a vertical plane at the front edge

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to a vertical plane at the most forward part of the seat back cushion and shall be constructed with full depth polyester or polyurethane foam. Individually constructed seat backs shall include headrests and shall be of full depth polyester or polyurethane foam with a free height of not less than 26 inches from the seat cushion. Seat grab rails and coatrails shall not be furnished. Cross tube snug with seat back is acceptable.

* 3.5.8.3 Passenger seating for types IV and V buses. All seats for types IV and V school age passenger buses, shall provide a 13-inch seat width per passenger. Seats shall be the manufacturer's standard forward facing type mounted on each side of the center aisle, and shall accommodate the number of school age passengers specified (see 1.2). Seats located at or near wheelhousings shall be installed to minimize interference with passenger comfort. Seat spacing shall be the maximum allowed by FMVSS 222. Spacing at padded barrier or other obstructions at the door entrance shall be the maximum allowed by FMVSS 222.

* 3.5.8.3.1 School age passenger seat construction. All school age passenger seat frames shall be of formed metal tube construction. Seats shall be the manufacturer's standard 39 inch width, three-passenger type with a seat cushion free depth of not less than 15 inches. Width of rearmost seats and of front seat at entrance may be reduced to nominal 26 inches, if required to provide adequate access to entrance and emergency exit. Seat cushion free depth shall be measured from a vertical plane at the front edge of cushion to a vertical plane at the most forward part of the seat back cushion. Seat cushions shall be solid foam padding tapering from a minimum of 5 inches at the front to 3 inches at the rear or not less than 2 inches of padding on a spring base. Seat backs shall be shaped and padded to assure passenger comfort. Seat backs shall be not less than 18 inches free height from seat cushion to top of seat back cushion.

* 3.5.8.4 Driver's seat. Driver's seat shall be heavy duty bus-type, pedestal mounted and adjustable fore, aft, and for height, without the use of tools. When a front mounted engine is furnished, the driver's seat shall also be adjustable sideways; and with the seat in the extreme left side slide position, there shall be a clear pass through of not less than 7-1/2 inches between the seat and engine housing. The seat shall be positioned for convenient and safe operation of all controls by the driver. Seat cushion and back shall be padded with full depth foam rubber, contoured to provide maximum driver comfort, and covered with heavy duty vinyl simulated leather. A seat belt with automatic retractors shall be provided for the driver's seat.

3.5.9 Stanchions and grab rails. One padded stanchion shall be located near the driver with a header rail and a slide or roll type opaque curtain and a padded cross rail to protect the driver from passenger interference. A padded stanchion or padded grab rail with a

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modesty panel shall be provided at the rear of the service door entrance. A stanchion or grab rail at the front of the service doors arranged to safely assist passengers entering or leaving the bus shall be provided. Full length grab rails shall be attached to the ceiling on each side of the aisle extending from at least the rear face of the first two-passenger seat to a point within 24 inches of the face of the rear seats. The stanchion or grab rail, header rail and cross rail located behind the driver and at the service door and the ceiling grab rails shall be of stainless steel, stainless steel clad, or plastic-coated steel tubing, at least 1 inch in diameter.

3.5.10 Heating, defrosting and ventilation. The heating, defrosting and ventilating system shall conform to Federal Motor Carrier Safety Regulations 393.77 and 393.79.

3.5.10.1 Heater. A hot-water heating system having a total capacity of not less than the number of British thermal units (Btu) per hour specified in table II at 150°F water to ambient air temperature differential, shall be installed for the type bus furnished. The bus heating system shall be capable of raising the interior temperature of the bus from 0°F to 50°F in 20 minutes (with bus soaked in an ambient temperature of 0°F for 15 hours and with no more than a 10 minute engine warmup time). Temperature shall be measured 36 inches inward from the side walls, windshield and rear walls. Heat shall be evenly distributed through the aisle area. A booster pump shall be provided for proper water circulation. Adjustable manual controls shall be provided to permit outside air, recirculated air, or variable mixtures as desired, to circulate through at least the main heater core. Heater(s) shall be located to assure flow of heated air to driver's area, to all parts of the passenger compartment, and so that no objectionable flow is directed toward any passenger. Heater hoses shall be shielded from the passenger compartment and located in protected positions. When specified (see 6.2), not less than four heaters with booster pump(s), two front, one middle, and one to the rear, shall be furnished.

TABLE II. Heater requirements, system.

Type bus	Number of Btu
I	110,000
II	120,000
III, IV	130,000
V	150,000

3.5.10.2 Defroster. Defroster outlets with suitable connections and independent blower(s) shall be furnished to provide a flow of heated air to each side or section of the windshield. The windshield defrosting system shall conform to SAE J381 and SAE J382.

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3.5.10.3 Ventilator. Except when air conditioning is furnished (see 3.5.14), four power ventilators shall be located in the roof, one over each wheelhousing. Each blower motor shall be capable of moving not less than 185 cubic feet of air per minute and shall have not less than two speeds controlled by separate switches mounted in a location easily accessible to the driver. Outside openings for heaters and ventilators shall be so located as to minimize entry of fumes, rain, road dust and road wash. Ventilator openings shall be provided with a weatherproof cover with means for opening and closing from inside the bus.

* 3.5.11 Body equipment. Body equipment shall be complete with all accessories listed by the body manufacturer as standard equipment and, in addition, all buses shall include the items specified in 3.5.11.1 through 3.5.11.10.

3.5.11.1 Rearview mirrors. One nonglare type mirror shall be mounted inside the bus and two mirrors outside, one each at left- and right-hand sides, mounted to permit maximum vision without obstruction, and visible on right through the upper corner of the windshield, and on the left through driver's window(s). Outside rearview mirrors shall be of the combination (flat and convex) type. Each combination mirror shall have at least 50 square inches of flat reflective area and a convex surface having at least 20 square inches of reflective area. Inside rearview mirror shall have a reflective surface area of not less than 80 square inches.

3.5.11.1.1 Front cross view mirror. When specified (see 6.2), a manufacturer's standard front cross view mirror shall be provided at the right front of the bus to provide driver view of the front bumper and the area in front of the bus.

3.5.11.2 Sunvisor. An adjustable sunvisor, not less than 6x16 inches, shall be furnished.

3.5.11.3 Emergency reflective triangles. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i) shall be furnished and properly stowed.

* 3.5.11.4 Emergency equipment. A 5-pound dry chemical type fire extinguisher conforming to Federal Motor Carrier Safety Regulation 393.95(a) shall be mounted in a location convenient to the driver. The fire extinguisher shall have an Underwriter's Laboratory rating of 10B:C or more, shall include a metal safety pin and wire and shall conform to type I or type II, class 2, size 5 of A-A-393. The fire extinguisher shall permit visual determination of full charge.

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3.5.11.5 Stowage compartment(s). Provision shall be made for weatherproof stowage of tools and tire chains, when carried, to be secured by padlock. A padlock is not required.

3.5.11.6 Rear bumper. Rear bumper shall be heavy steel channel type with antiride and antihitch features attached in a manner to permit the bus to be pushed without permanent distortion of bumper, body or chassis parts.

3.5.11.7 Keys. Duplicate keys shall be furnished for all key-operated locks, and properly identified keys shall be securely packed and attached to the vehicle steering wheel or otherwise stowed in an approved manner for shipment. All locks on the same vehicle shall be of the same key code, except the ignition key.

3.5.11.8 Signs. Signs to read "DANGER - DO NOT EXTEND ARMS FROM WINDOWS" shall be affixed to the interior of the body. Signs shall be legible at a distance of 20 feet and shall be located as follows: one at the front, one at the rear, and one on each side.

3.5.11.9 Standee line. Areas prohibited to standees shall be marked in accordance with Federal Motor Carrier Safety Regulation 393.90.

3.5.11.10 Baggage compartment(s) (types I, II and III only). Unless otherwise specified (see 6.2), types I, II and III adult passenger buses shall be provided with weatherproof baggage compartment(s) under the floor. The baggage compartments shall be illuminated, shall be carpeted with vinyl or rubber mat flooring, and shall be of the largest capacity available. Compartment(s) shall be provided with key operated locking device(s). Total capacity of the baggage compartment(s) shall be not less than 16 cubic feet for type I buses with and without air conditioning, 40 cubic feet for type II buses without air conditioning, 20 cubic feet for type II buses with air conditioning, 42 cubic feet for type III buses without air conditioning, and 25 cubic feet for type III buses with air conditioning. For type I bus with air conditioning when sufficient below floor baggage space is not available, a baggage space shall be provided behind the rearmost seats to provide a total capacity above and below floor of not less than 16 cubic feet. Above floor baggage space shall be considered bounded by the seat height and width.

3.5.12 Body mounting. Heavy rubber cushions or antisqueak material shall be utilized between the body frame and attaching chassis parts. Reinforcements or filler blocks shall be used where mounting devices might otherwise deform frame flanges. Mounting devices shall be locked units which will minimize loosening, but which may be tightened if necessary.

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* 3.5.13 Air conditioning. When specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard independent diesel engine driven air conditioning system conforming to 3.5.13.1 through 3.5.13.2.6. Manufacturer shall furnish to the engineering department of the procuring activity the rated capacity at the specified conditions of the entire unit, either by the air measurement or the flow meter method.

3.5.13.1 Performance. The air conditioner shall have a minimum net cooling capacity of 48,000 Btu per hour for types I and II buses and 68,000 Btu per hour for types III IV and V buses in a 95°F ambient temperature; and at this temperature and in all conditions of solar radiation, system shall be capable of maintaining not more than 80°F dry bulb and a 68°F wet bulb and an effective temperature of 74°F in the bus. The humidity shall be not less than 30 percent, and not more than 60 percent. The inside temperature shall be measured 6 inches from the floor, 6 inches from the ceiling, and at seat level. Longitudinal measurement shall be taken at the front seat, at the rear seat, and midway of the passenger area. The inside temperature gradient within the bus shall be not more than 6°F. The cooling performance specified herein shall be met when the bus is traveling at 50 mph and when standing still. The specified air conditioner minimum net cooling capacity in Btu per hour shall be increased, if required, to meet the performance requirements specified herein. The air conditioning system shall be independent of the heating system and shall automatically proportion air in the ratio of 20 percent fresh air to 80 percent recirculated air. All air shall pass through the filter system.

* 3.5.13.2 Components. The air conditioning system shall be powered by an auxiliary diesel engine of sufficient horsepower so that when fully loaded at sea level conditions it shall not utilize more than 65 percent of available power output under constant load conditions. Air conditioning power source and compressor shall be located in a compartment under the floor, in a compartment at the rear, or on the roof and shall be accessible for ease of maintenance and service. The compartment door shall be capable of being opened and secured closed without the use of tools. When the compressor and engine are roof mounted, an access ladder shall be provided. The condenser and evaporator may be mounted overhead on the roof. Air conditioner controls shall be panel mounted within easy reach of the seated driver. A second set of controls shall be located in the auxiliary engine compartment in series with driver controls and with safety switch to cut out driver controls. Compartment lamp switch and junction panel shall be installed in the auxiliary engine compartment. Thermostat control shall be located in the return air stream. The air conditioner shall be designed for bus operation and shall be complete with all necessary controls for automatic operation of the unit after power unit has been started. The compressor shall be controlled automatically on demand

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of the thermostat. Mountings shall be so as to minimuze transfer of vibration. The size and weight of the assembly shall be held to a minimum consistent with the performance requirements specified herein.

* 3.5.13.2.1 Engine components. The engine shall be a two- or four-cycle engine equipped with at least the following components:

(a) Self-starter, 12-volt.
(b) Air cleaner.

3.5.13.2.2 Governor. A variable-speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed.

3.5.13.2.3 Electrical system. The electrical system shall utilize the bus chassis batteries and shall conform to Federal Motor Carrier Safety Regulations 393.30 through 393.33.

3.5.13.2.4 Fuel pump. A fuel pump shall be furnished and when lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided, either as a component of the fuel pump or between the fuel pump and the carburetor.

3.5.13.2.5 Exhaust system. The exhaust system shall be furnished with spark arresting muffler, shall conform to 3.4.4, and shall exhaust fumes away from the body to the rear of the rear wheels.

* 3.5.13.2.6 Auxiliary engine fuel tank. The auxiliary engine may be supplied from the main engine tank(s) or from a separate auxiliary fuel tank. When a separate auxiliary engine fuel tank is furnished, it shall have a capacity of not less than 20 gallons and shall be located under the floor. The tank shall be a safety fuel tank and of welded seam construction. If the tank extends below the main body frame rails, the tank material shall be not less than 12 gage (0.1046 inch) thickness or the tank shall be protected at front, back and bottom with a steel protection plate of not less than a 12 gage (0.1046 inch) thickness. A drain shall be provided in the fuel tank sump, located to avoid drainage of fuel on any part of the vehicle. Fuel tank filler neck shall be located on the side of the bus with access through flush type door. A photoetched, corrosion-resistant plate, permanently protected decal or stencil, shall be located in a conspicuous place near the filler door and shall state "Diesel fuel-auxiliary engine." A shutoff valve shall be provided in the fuel line adjacent to the tank. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67. When a separate auxiliary fuel tank is furnished, a separate fuel gage shall be provided.

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3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government, the contractor shall service and adjust the vehicle for operational use including at least the following: alinement of lights; adjustment of engine and brake systems; filling and charging of the batteries; alinement of front wheels; inflation of all tires; complete lubrication of body, chassis, engine and running gear, with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

* 3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry. All hoses, wires and rods shall be supported and restrained where necessary, to minimize vibration and to prevent damage. The supports and restraints shall not interfere with either normal or emergency operations. Sharp edges and corners shall be either padded or rounded to prevent injury to personnel.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The supplier shall provide the Government inspection representative instruments and such assistance as the representative may find necessary.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification

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at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle of each type produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 4.3.2 Noise, heating, and air conditioning test. The vehicle shall be tested to determine conformance to all noise level, heating, and air conditioning performance requirements specified in 3.5.2.1, 3.5.10.1, and 3.5.13.1 respectively. Tests shall be as specified in 4.3.2.1, 4.3.2.2, and 4.3.2.3. Test data, including a description of all applicable and pertinent test conditions, shall be recorded and furnished to the Government representatives. If the supplier has previously tested an identical model vehicle with the same components, a certification and the specified test data verifying conformance to this specification may be provided in lieu of first vehicle tests. Certification and previous test results, if furnished, shall be forwarded to the engineering office of the procuring activity (USATACOM, AMSTA-UED).

4.3.2.1 Noise level. Noise level shall be measured at seated ear height, with all windows closed, at various seat locations, and at designated vehicle speeds. A reading shall be taken at each location and speed specified (see 3.5.2.1). At 30 mph, the readings taken at the three positions shall be averaged to determine conformance to column A of table I. At the authorized maximum speed of the vehicle, the maximum reading at each position shall be taken to determine conformance to column B of table I. Sound measuring system shall consist of a condenser microphone, a sound level meter, and an octave band analyzer which meet or exceed the instrumentation specified in MIL-STD-1474 (Steady-State Noise, Personnel-Occupied Areas). The sound measuring system shall be calibrated before the test and at the conclusion of the test.

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4.3.2.2 Heating, defrosting and ventilating. To verify conformance to the heating requirements specified in 3.5.10.1, the vehicle manufacturer shall test the vehicle under actual or simulated conditions at least as stringent as specified herein, or shall furnish the engineering calculations and certification of conformance to the performance and required Btu capacity of the bus heating system. The engineering calculations shall include:

(a) Estimated engine heat output in Btu/hr and water temperature.
(b) Total heater output in Btu/hr and water to ambient temperature differential.
(c) Body assembly heat loss in Btu/hr.
(d) Estimated time required to raise the interior temperature from 0°F to 50°F (when soaked in an ambient temperature of 0°F for 15 hours) when 170°F hot water is applied at the rate of three gallons per minute. A 10-minute engine pre-warmup time can be assumed.
(e) No heat gain from solar radiation and passenger body heat.
(f) Any significant heat losses.

Heating, defrosting, and ventilation systems and controls shall be operated to determine if they are functioning properly.

4.3.2.3 Air conditioning. To verify conformance to the air conditioning performance requirements specified in 3.5.13.1, the manufacturer shall test the vehicle under actual or simulated conditions at least as stringent as specified herein. Simulated conditions shall be acceptable to the Government. Inside temperature may be calculated either with the specified number of seated passengers, or the use of simulated heat, for sensible cooling. Solar radiation may be employed or simulated by calculating solar load effective in Btu, based on the square foot surface and the angle of incidence that will be exposed to the sun at the same time.

* 4.3.3 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on the front and rear axles. The total imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds, and the payload required to provided the specified GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

* 4.3.4 Road test. The first production vehicle of the largest capacity adult passenger type and of the largest capacity school age

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passenger type furnished under the contract shall be examined and road tested with payload by the contractor to assure that the vehicle(s) will operate in accordance with the contract requirements.

* 4.3.4.1 Road test details. Unless otherwise specified (see 6.2), the vehicle(s) shall be loaded to simulate maximum payload and shall be subjected to a road test of not less than the mileage specified in table III. The test mileage shall be as indicated for the total quantity of vehicles being procured under this specification under the contract at the time of contract award. Test shall be conducted by the contractor under the direction of Government representatives. Test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table IV. Vehicle(s) shall be inspected periodically during and at completion of test, by the contractor. The cost of test and vehicle support, exclusive of personal expenses incurred by participating Government representatives, shall be borne by the contractor.

4.3.4.1.1 Test course. The total test mileage shall be not less than the mileage specified in table III on roads and highways within the general geographic limits of the place of manufacture and adjoining states. The course shall be arranged to return the vehicle to the manufacturer's plant for inspection and chassis servicing at the completion of each 1000 miles and at the end of the test, if less than a 1,000 mile increment is required under table III. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall preclude recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries.

TABLE III. Road test.

Vehicle contract quantity *	Total test mileage
1- 5	250
6-10	1,000
11-20	2,500
21 or more	5,000

* NOTE: The vehicle contract quantity shall be the total number of adult passenger (types I, II amd III) and the total number of school age passenger (types IV and V) buses, as applicable. These total quantities shall determine the road test mileage for the bus(es) (see 4.3.4).

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TABLE IV. Test course requirements.

Type of road	Percent of minimum miles
Dirt	2
Gravel	18
City streets	10
Unimproved black top	20
Improved pavement	50

4.3.4.1.2 Test failure. The vehicle utilized for the road test shall successfully complete the test. Substitution of new vehicle or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test vehicle shall be for deficiencies, including but not limited to the following:

(a) Damage caused by collision.
(b) Failure of any major component.
(c) Vibration due to misalinement of wheels, frame, or driveshaft.
(d) Vibration due to type of body construction or mounting.
(e) Failure of any driveline component.
(f) Evidence of abnormal tire wear due to alinement or imbalance.
(g) Failure of any vehicular safety device such as brakes, steering assembly, windshield washers and wipers, or electrical circuits.
(h) Evidence of structural weakness in any part of vehicle, vehicle components or accessories.
(i) Loose mounting of parts or accessories due to workmanship or vehicular operation.
(j) Excessive oil consumption (exceeding 1 quart per 1000 miles).

4.3.4.1.3 Mileage and maintenance records. The vehicle shall be driven not more than a total of 1,000 miles over the miles required under table III, including break-in mileage, prior to and during the test. All mileage accumulated and all service or maintenance performed during test shall be recorded and copies of service and maintenance records shall be furnished to the Government representatives. Distribution of records shall be identical to that for service manuals and shall be concurrent with the manual distribution.

4.3.4.1.4 Delivery of test vehicle. Upon successful completion of vehicle test and final examination, the vehicle shall be cleaned, serviced and refurbished to the extent required for delivery of a new vehicle to the Government. Warranty provisions of the contract shall be exclusive of total test miles.

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* 4.3.5 Production sample. When the first production vehicle has been accepted it shall be refurbished and it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. Each production vehicle with all components, less payload, shall be examined and vehicle driven on approximately level road surfaces by the contractor to assure that the vehicle will operate in accordance with the contractual requirements. The contractor may authorize his sub-contractor who will perform the final assembly work on the vehicle to make such examination and tests, but this shall not relieve the contractor of his obligation to deliver an acceptable vehicle or constitute a waiver of any warranty provision. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 has been accomplished.

4.4.1 Removal of instruments and controls. The requirements in 3.4.17 shall not be construed as requiring the removal of the instruments and controls as part of the final examination and tests. However, the presentation of the complete vehicle with said instruments and controls in place shall be tantamount to certification that said instruments and controls are installed in a manner that facilitate their removal and servicing.

4.4.2 Rejection. Deficiencies of workmanship and nonconformance to any requirements of the contract shall be cause for rejection until corrective action has been taken.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government in transporting both children and adults over roads and highways encountered by the Military services in the continental United States and overseas areas, and for operation in all climates. The principal use of the vehicles will be stop-and-go, short haul, high, medium, or low speed operation.

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* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and number of passengers (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed markings, if required (see 3.1.1.1).
(e) Gloss yellow exterior color, if required (see 3.1.1.1).
(f) Rustproofing, if required (see 3.1.1.2).
(g) If towing brakes are not required (see 3.1.1.9).
(h) Exterior sound level in accordance with the California Vehicle Code, if required (see 3.2.3).
(i) Power plant heaters and fuel warmer, if required (see 3.4.1.5).
(j) If a battery heater is not required (see 3.4.1.5).
(k) Silicone rubber radiator and heater hoses, if required (see 3.4.1.6).
(l) A fuel fired engine preheater, if required (see 3.4.1.7).
(m) Passenger signal buzzer, if required (see 3.4.2.6.2).
(n) If school bus signal lamps are not required (see 3.4.2.6.4).
(o) Master electrical switch, if required (see 3.4.2.7).
(p) Air cleaner aspirator, if required (see 3.4.3.1).
(q) Transmission lockup, if required for bus with automatic transmission (see 3.4.5.1).
(r) Manual transmission, if required (see 3.4.5.2).
(s) Traction control, if required (see 3.4.9.1).
(t) Disc type wheels, if required (see 3.4.10).
(u) If other than highway tread is required (see 3.4.10.1).
(v) Bias ply tires, if required (see 3.4.10.1).
(w) Spare wheel or rim and carrier, if required (see 3.4.10.4).
(x) Spare tire assembly, if required (see 3.4.10.5).
(y) Air operated horn, if required (see 3.4.12).
(z) Tools, if required (see 3.4.16).
(aa) Right hand drive, if required (see 3.4.17).
(ab) Destination sign compartment, if required (see 3.5.4.4).
(ac) Padlocking hasp on service door and locking bolt on emergency door, if required (see 3.5.5).
(ad) Four heaters, if required (see 3.5.10.1).
(ae) Front crossview mirror, if required (see 3.5.11.1).
(af) If baggage compartment is not required for adult passenger buses (see 3.5.11.10).
(ag) Air conditioning, if required (see 3.5.13).
(ah) Road test mileage, if different (see 4.3.4.1).

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6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirement for each vehicle model furnished under contract should be submitted as specified in the contract. Unless other conditions are cited in this contract, computations should be made for normal atmospheric pressure, normal ambient air temperatures, and still, dry air. The factors to be used in predicting vehicle ability (see 3.3.1.1) are established as follows for the corresponding SAE J688, Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.841
Table 4	- Area Factor. 0.197
Table 5	- Velocity Factor. 333
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW and the engine revolutions per minute (rpm) (to nearest 100 revolutions) which is required for a geared speed of 55 mph. For engine speed beyond the range of this table, the following formula shall be used: FHP = 1 + (0.0000002)(rpm)(GVW).
Table 8	- Grade Factor. 0.68
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

6.4 Subject term (key word) listing.

Bus
Bus, air conditioning system
Bus, brake system
Bus, forward control type
Bus, gradeability
Bus, heating system
Bus, performance
Bus, storage battery
Bus, suspension system

6.5 Identification of changes. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no

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liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

6.6 Classification change. Forward control model, 64 and 82 school-age passenger buses of MIL-B-62277F dated 1 November 1984, may be procured under type IV, 64 school-age passenger and type V, 82 school-age passenger of this specification.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Review activities:
Army - EL
Navy - MC

Preparing activity:
Army - AT

Project No. 2310-0435

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